

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-2023

Program: B Tech Integrated Stream: Computer Science

Subject: Chemistry-II

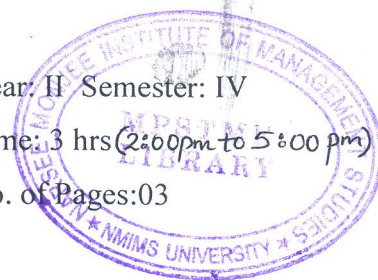
Date: 29 / 6 / 23

Marks: 100

Year: II Semester: IV

Time: 3 hrs (2:00 pm to 5:00 pm)

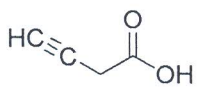
No. of Pages: 03



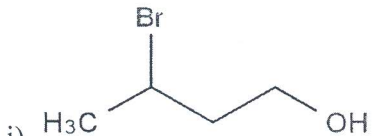
Re - Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**
- 8) **Atomic weights:** H=1, C=12, O=16, Al=27, N=14, Li = 6.9, Na=23, Mg=24, P=31, S=32, Cl=35.5, Ca=40, Ni=58.7, Fe=55.85, K=39, B=10.8

Q1		Answer briefly:	[20]
CO-1; SO-1 BL-2	a.	Explain electronegativity. Give the trend in the electronegativity of the following elements of period-2 in the descending order with proper justification: Be (Z=4), C (Z=6), F (Z=9), O (Z=8).	[5]
CO-2 ; SO-1 BL-1	b.	Define and explain with formula the concept of molecular weight. Calculate the number of sodium atoms in a sample of 636g Na ₂ CO ₃ .	[5]
CO-3; SO-1 BL-3	c.	Identify functional groups and write IUPAC names of following compounds: i) CH ₂ =HC-CH ₂ -CH ₂ -CHO ii) 	[5]
CO-4; SO-1 BL-1	d.	Define osmosis, reverse osmosis and osmotic pressure. What would happen to the osmotic pressure if excess solute is added to the solvent?	[5]
Q2 CO-1; SO-1 BL-1	a.	What do you understand by organic chemistry? Discuss tetravalency of carbon with the help of electronic configuration.	[5]

CO-2; SO-1 BL-3	b.	How many moles of nitrate ions are in a mixture of 2 mol of calcium nitrate $\text{Ca}(\text{NO}_3)_2$ and 1.5 mol of aluminium nitrate $\text{Al}(\text{NO}_3)_3$?	[5]
CO-1; SO-1 BL-1	c.	Define effective nuclear charge. Estimate Z_{eff} of 4s electron for Zn ($Z=30$) and 3p electron in Si (14)	[5]
CO-2; SO-1 BL-3,5	d.	Calculate the weight of a sample of nickel sulphate hexahydrate ($\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$) that contains 69×10^{30} atoms.	[5]
Q3 CO-3; SO-1 BL-1	a.	Explain carboxylic acid functional group with respect to identification, 2 methods of preparation and 2 reactions.	[5]
CO-2; SO-1 BL-3	b.	What mass of Fe is present in Fe_3O_4 that contains 4.8×10^{23} atoms of oxygen?	[5]
CO-3; SO-1 BL-1	c.	Discuss co-ordinate bond with the help of NH_4^+ formation.	[5]
CO-4; SO-1 BL-3	d.	A student prepared a solution by dissolving 23 g NaCl in water such that total volume of solution is 750 ml. Density of solution is 1.09 g/ml. Help him calculate molality of solution.	[5]
Q4 CO-3; SO-1 BL-1	a.	Define functional group and give examples with structures of any 4 functional groups	[5]
CO-3; SO-1 BL-1,2	b.	$\text{CH}_3 - \underset{\text{1}}{\overset{\text{OH}}{\text{C}}} = \text{O} \xrightarrow{\text{R1}} \text{CH}_3 - \underset{\text{2}}{\overset{\text{H}}{\text{C}}} = \text{O} \xrightarrow{\text{R2}} \text{CH}_3 - \underset{\text{3}}{\text{CH}_2\text{OH}}$ In the above reaction : - identify functional groups of 1, 2 and 3 - Names of 1,2 and 3 - type of reaction R1 and R2	[5]
CO-3; SO-1 BL-4	c.	Illustrate the structure of the following compounds: - m-chloro benzaldehyde 3-methyl-2-butan-2-one 1,3-Dichlorobenzene 1-bromo-1,2-dichloropropane 2,5-dimethyl-1,5-hexadiene	[5]
CO-3; SO-1 BL-2	d.	Discuss SP^2 hybridization of ethene ($\text{CH}_2 = \text{CH}_2$).	[5]

Q5 CO-1; SO-1 BL-4	a.	Define the term ionization energy and mention periodicity across the period. Predict period, group and block of elements having atomic number 10 and 17.	[5]
CO-2; SO-1 BL-1,4	b.	Define the terms normality and equivalent weight. Calculate gram equivalents of 3.52g H ₃ PO ₄ and 2.73g of LiOH	[5]
CO-1; SO-1 BL-1,4	c.	What should be the electronegativity difference between atoms, for a bond to be polar, non polar and ionic? A bond is formed between H and O. Electronegativity of O is 3.44 and H is 2.1. Comment and justify if the bond formed between O and H is non-polar covalent or polar covalent or ionic bond.	[5]
CO-2; SO-1 BL-3	d.	Define solution, solute and solvent. How are solutions classified on the basis of gas as a solvent? Give suitable examples	[5]
Q6 CO-3; SO-1 BL-1,4	a.	Explain the formation of ionic and polar covalent bond with an appropriate example(s).	[5]
CO-4; SO-1 BL-4	b.	A solution was prepared by dissolving 42 g KCl in water such that total volume of solution is 300 mL. Density of solution is 0.99 g/mL. Calculate mole fraction of both the components of solution.	[5]
CO-3; SO-1 BL-2	c.	Explain alcohol functional group with respect to identification, 2 methods of preparation and 2 reactions.	[5]
CO-1; SO-1 BL-1	d.	What are isoelectronic species? With examples discuss the size trends of isoelectronic cations and anions .	[5]
Q7 CO-3; SO-1 BL-2	a.	Discuss aldehyde & ketone functional group with respect to 1 identification test, 2 method of preparation and 2 reaction of each.	[5]
CO-3; SO-1 BL-2	b.	What do you understand by octet rule? Explain any 2 exceptions of octet rule.	[5]
CO-3; SO-1 BL-2	c.	Predict IUPAC names of following compounds:  i) <chem>CC(Br)CCCO</chem> ii) <chem>CH3-CH=CH-CH=CH-CH3</chem>	[5]
CO-2; SO-1 BL-3	d.	Show Lewis dot structure of BCl ₃ and N ₂ . Estimate total number of atoms in 60 g BCl ₃ .	[5]